

## MIDDLESBORO-BELL COUNTY AP, KY, USA

WMO: 720353

Lat: 36.611N

Lon: 83.738W

Elev: 1154

StdP: 14.09

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 63875

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 13.6       | 18.6 | 0.7                           | 5.9 | 19.5 | 6.3 | 7.9 | 24.0 | 17.2                  | 46.9 | 15.6 | 44.0 | 2.9                   | 340  | 0.373 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.2                   | 92.8            | 75.8 | 90.6 | 74.6 | 88.8 | 73.6 | 79.5                | 87.5 | 77.6 | 86.6 | 76.0 | 84.7 | 3.8                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.4                            | 148.8 | 83.1 | 75.1 | 137.4 | 81.4 | 73.3 | 129.2 | 79.4 | 44.0          | 88.3 | 42.0 | 85.8 | 40.3 | 85.4 | 85.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| 14.1              | 11.8 | 10.3 | 8.1                        | 95.3 | 6.1                | 4.6 | 3.7                                                | 98.5 | 0.1        | 101.2 | -3.3       | 103.8 | -7.8       | 107.1 |    |    |
|                   |      |      | 5.2                        | 79.5 | 7.4                | 2.6 | -0.1                                               | 81.4 | -4.5       | 83.0  | -8.6       | 84.4  | -14.0      | 86.3  |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 58.3   | 37.3  | 40.7  | 48.9  | 58.6  | 66.8  | 74.1  | 76.6  | 76.2  | 70.9  | 59.0  | 47.7  | 41.2 |
|                                                                               | DBStd       | 16.23  | 10.91 | 10.66 | 10.08 | 8.45  | 7.09  | 4.74  | 4.54  | 4.34  | 5.68  | 8.53  | 8.50  | 9.13 |
|                                                                               | HDD50       | 1342   | 413   | 296   | 146   | 23    | 1     | 0     | 0     | 0     | 0     | 20    | 140   | 303  |
|                                                                               | HDD65       | 3842   | 858   | 682   | 506   | 223   | 69    | 2     | 0     | 0     | 17    | 224   | 522   | 739  |
|                                                                               | CDD50       | 4351   | 20    | 34    | 111   | 280   | 523   | 723   | 826   | 812   | 627   | 297   | 69    | 29   |
|                                                                               | CDD65       | 1379   | 0     | 0     | 6     | 30    | 125   | 276   | 361   | 348   | 194   | 37    | 2     | 0    |
|                                                                               | CDH74       | 12239  | 0     | 3     | 81    | 442   | 1218  | 2517  | 3117  | 2930  | 1565  | 353   | 13    | 0    |
|                                                                               | CDH80       | 4694   | 0     | 0     | 13    | 92    | 383   | 1004  | 1314  | 1225  | 589   | 74    | 0     | 0    |
| Wind                                                                          | WSAvg       | 2.7    | 3.6   | 3.8   | 3.9   | 3.8   | 2.6   | 2.1   | 1.7   | 1.5   | 1.5   | 2.1   | 2.5   | 3.0  |
| Precipitation                                                                 | PrecAvg     | 51.2   | 4.2   | 4.0   | 5.3   | 4.2   | 4.9   | 4.3   | 5.2   | 4.3   | 3.1   | 3.2   | 4.2   | 4.4  |
|                                                                               | PrecMax     | 66.7   | 10.3  | 7.1   | 12.2  | 9.6   | 9.8   | 9.1   | 12.7  | 11.1  | 5.7   | 6.2   | 10.9  | 9.2  |
|                                                                               | PrecMin     | 39.7   | 1.2   | 0.5   | 1.5   | 0.5   | 1.8   | 1.0   | 1.1   | 1.3   | 0.5   | 0.0   | 1.0   | 0.4  |
|                                                                               | PrecStd     | 7.2    | 2.1   | 1.7   | 2.6   | 1.9   | 2.0   | 1.9   | 2.9   | 2.2   | 1.4   | 1.7   | 2.1   | 2.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 66.3  | 72.5  | 81.7  | 87.7  | 90.4  | 93.5  | 97.2  | 95.4  | 91.5  | 85.9  | 75.6  | 69.3 |
|                                                                               |             | MCWB   | 60.1  | 61.6  | 63.9  | 64.6  | 72.7  | 74.8  | 78.5  | 77.9  | 72.1  | 70.6  | 62.3  | 62.1 |
|                                                                               | 2%          | DB     | 63.0  | 67.6  | 75.2  | 82.2  | 87.7  | 91.2  | 92.7  | 92.3  | 89.9  | 81.8  | 71.4  | 64.4 |
|                                                                               |             | MCWB   | 55.5  | 59.8  | 60.6  | 63.6  | 70.4  | 74.2  | 77.1  | 76.2  | 72.0  | 67.2  | 59.4  | 58.5 |
|                                                                               | 5%          | DB     | 58.8  | 63.4  | 71.5  | 79.3  | 84.2  | 89.6  | 90.1  | 90.1  | 87.0  | 78.7  | 66.5  | 61.2 |
|                                                                               |             | MCWB   | 53.2  | 53.8  | 58.4  | 61.7  | 68.6  | 73.5  | 75.7  | 74.9  | 72.0  | 66.1  | 57.3  | 56.6 |
|                                                                               | 10%         | DB     | 54.5  | 58.5  | 66.2  | 75.2  | 81.5  | 86.0  | 87.7  | 87.5  | 83.5  | 73.2  | 63.4  | 56.6 |
|                                                                               |             | MCWB   | 49.9  | 50.8  | 55.8  | 60.2  | 67.5  | 72.0  | 74.7  | 74.5  | 70.6  | 63.7  | 55.7  | 52.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 63.4  | 64.0  | 66.9  | 69.9  | 75.2  | 81.0  | 81.8  | 82.9  | 77.8  | 73.9  | 65.4  | 63.8 |
|                                                                               |             | MCDB   | 65.1  | 69.4  | 76.4  | 80.5  | 86.8  | 90.0  | 90.3  | 89.8  | 86.5  | 82.0  | 71.2  | 67.5 |
|                                                                               | 2%          | WB     | 57.4  | 60.8  | 63.7  | 66.6  | 72.8  | 77.5  | 79.7  | 79.2  | 75.5  | 70.8  | 62.8  | 60.1 |
|                                                                               |             | MCDB   | 61.4  | 66.5  | 70.9  | 76.4  | 82.8  | 86.0  | 88.5  | 85.8  | 83.6  | 77.4  | 67.7  | 63.6 |
|                                                                               | 5%          | WB     | 54.3  | 56.6  | 60.6  | 64.3  | 71.1  | 75.2  | 77.9  | 77.1  | 73.9  | 68.0  | 60.0  | 57.0 |
|                                                                               |             | MCDB   | 57.2  | 60.5  | 68.9  | 74.2  | 80.5  | 84.4  | 87.0  | 86.1  | 82.0  | 74.0  | 64.5  | 59.9 |
|                                                                               | 10%         | WB     | 50.8  | 52.6  | 57.7  | 62.5  | 69.3  | 73.4  | 76.2  | 75.5  | 72.3  | 65.6  | 56.9  | 53.2 |
|                                                                               |             | MCDB   | 53.9  | 57.4  | 64.8  | 72.1  | 77.9  | 82.1  | 84.4  | 84.3  | 79.7  | 71.9  | 62.2  | 56.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 18.3  | 19.5  | 22.4  | 24.8  | 23.1  | 22.2  | 20.2  | 20.6  | 21.8  | 23.3  | 23.2  | 17.5 |
|                                                                               |             | MCDBR  | 21.6  | 25.3  | 29.3  | 30.9  | 27.3  | 26.0  | 23.0  | 23.6  | 26.1  | 27.4  | 28.2  | 21.1 |
|                                                                               |             | MCWBR  | 15.8  | 16.4  | 15.8  | 14.2  | 11.7  | 11.3  | 9.7   | 10.3  | 11.2  | 13.6  | 16.9  | 15.4 |
|                                                                               | 5% WB       | MCDBR  | 17.2  | 20.2  | 24.4  | 24.1  | 23.5  | 22.0  | 20.3  | 21.1  | 21.9  | 22.1  | 20.3  | 19.0 |
|                                                                               |             | MCWBR  | 14.3  | 15.4  | 14.3  | 12.8  | 10.9  | 10.4  | 9.2   | 10.0  | 10.2  | 12.7  | 14.1  | 15.2 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.307  | 0.316 | 0.347 | 0.391 | 0.440 | 0.467 | 0.497 | 0.505 | 0.405 | 0.364 | 0.324 | 0.305 |      |
|                                                                               | taud        | 2.506  | 2.474 | 2.422 | 2.302 | 2.181 | 2.140 | 2.089 | 2.046 | 2.311 | 2.421 | 2.501 | 2.530 |      |
|                                                                               | Ebn at Noon | 283    | 293   | 292   | 283   | 269   | 261   | 252   | 246   | 268   | 271   | 273   | 276   |      |
|                                                                               | Edh at Noon | 25     | 30    | 34    | 41    | 47    | 49    | 51    | 52    | 38    | 31    | 25    | 23    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 674    | 861   | 1230  | 1611  | 1790  | 1944  | 1811  | 1720  | 1455  | 1129  | 814   | 589   |      |
|                                                                               | RadStd      | 62     | 115   | 112   | 112   | 144   | 113   | 105   | 89    | 137   | 145   | 92    | 60    |      |

## Historical Trends

|      | DBAvg                   | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (22 neighbors) | +0.92   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +250  | +146  |

Nomenclature: See separate page